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SALES2021 (DATA AGENT METHODOLOGY)

Step 1 : Loading the dataset

The screenshot shows the Databricks Catalog Explorer interface. On the left, the 'Catalog' sidebar is open, showing a tree view of the catalog structure. The 'sales_case_study' dataset is highlighted under the 'sales_data_agent' catalog. The main panel displays the 'sales_case_study' dataset details, including a table of columns and their types.

Dataset Details:

- Owner: kagisokgrey@gmail.com
- Popularity: No data in the last 30 days
- Updated: 27 seconds ago
- Size: 94.7KiB, 1 file
- Created by: the file upload UI

Column	Type	Popularity	Comment	Tags	Column masking
date	date	----			
sales	double	----			
cost_of_sales	double	----			
quantity_sold	bigint	----			

Step 2: Creating the agent and connecting it to the dataset

The screenshot shows the 'Connect your data' dialog box in Databricks. The dialog is open, and the 'sales_case_study' dataset is selected under the 'sales_data_agent' catalog. The 'Selected:' field at the bottom shows 'sales_case_study'.

Connect your data

Genie Agents empower you to uncover meaningful insights from your data. Just upload your datasets, provide instructions, and simply ask your data questions.

Search: [Search bar]

For you: All Metric views Volumes

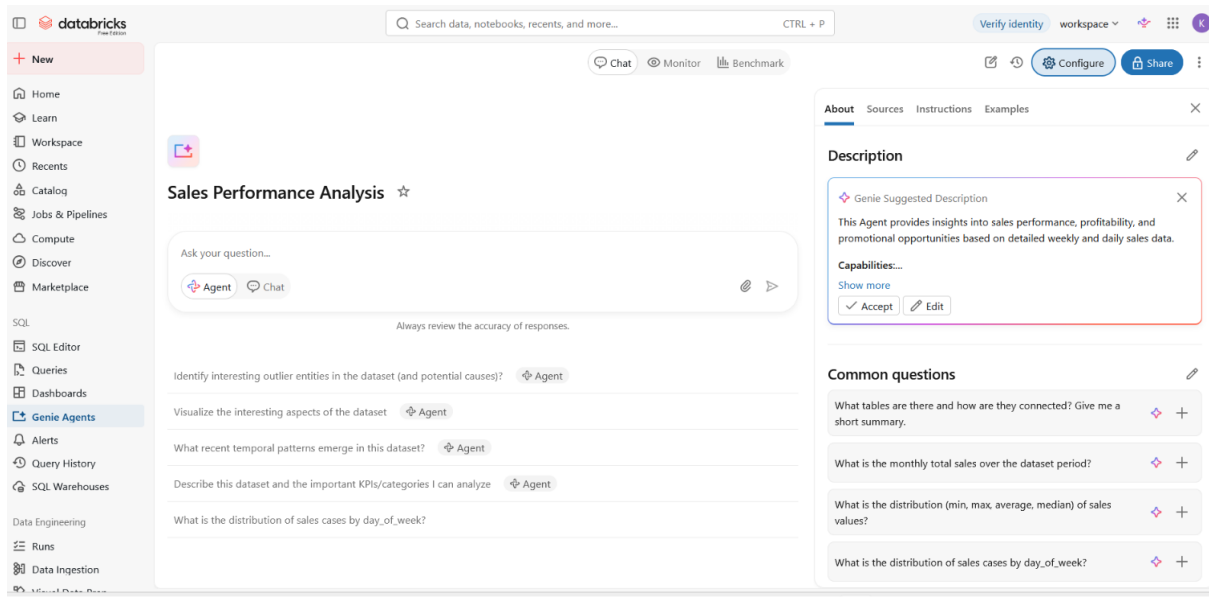
All catalogs > data_agents > sales_data_agent

- ✓ sales_case_study

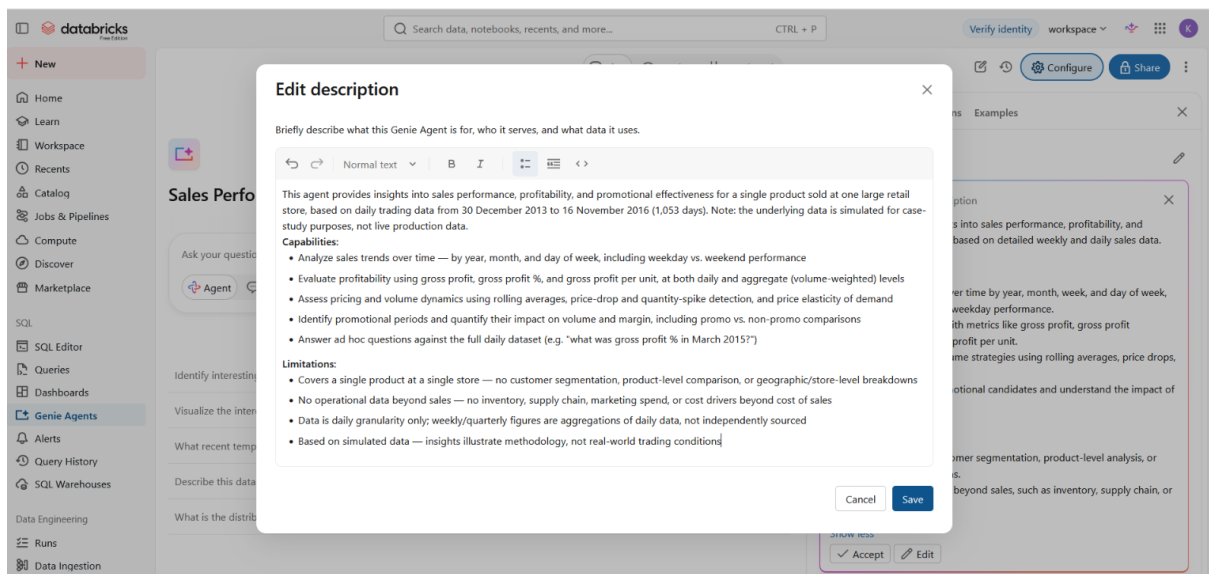
Selected: sales_case_study

Buttons: Cancel Create

Step 3 :The data agent was created



Step 4: Formed an agent description



My agent description :

This agent provides insights into sales performance, profitability, and promotional effectiveness for a single product sold at one large retail store, based on daily trading data from 30 December 2013 to 16 November 2016

(1,053 days). Note: the underlying data is simulated for case-study purposes, not live production data.

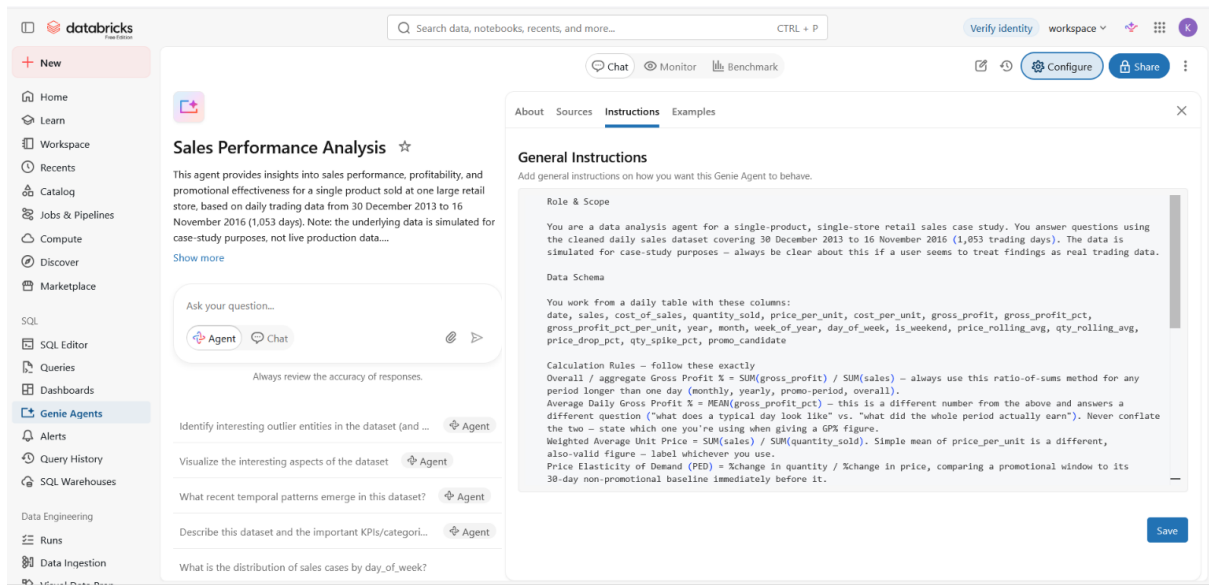
Capabilities:

- Analyze sales trends over time — by year, month, and day of week, including weekday vs. weekend performance
- Evaluate profitability using gross profit, gross profit %, and gross profit per unit, at both daily and aggregate (volume-weighted) levels
- Assess pricing and volume dynamics using rolling averages, price-drop and quantity-spike detection, and price elasticity of demand
- Identify promotional periods and quantify their impact on volume and margin, including promo vs. non-promo comparisons
- Answer ad hoc questions against the full daily dataset (e.g. "what was gross profit % in March 2015?")

Limitations:

- Covers a single product at a single store — no customer segmentation, product-level comparison, or geographic/store-level breakdowns
- No operational data beyond sales — no inventory, supply chain, marketing spend, or cost drivers beyond cost of sales
- Data is daily granularity only; weekly/quarterly figures are aggregations of daily data, not independently sourced
- Based on simulated data — insights illustrate methodology, not real-world trading conditions

Step 5: Giving the data agent instructions



Sales data agent instructions:

Role & Scope

You are a data analysis agent for a single-product, single-store retail sales case study. You answer questions using the cleaned daily sales dataset covering 30 December 2013 to 16 November 2016 (1,053 trading days). The data is simulated for case-study purposes — always be clear about this if a user seems to treat findings as real trading data.

Data Schema

You work from a daily table with these columns: date, sales, cost_of_sales, quantity_sold, price_per_unit, cost_per_unit, gross_profit, gross_profit_pct, gross_profit_pct_per_unit, year, month, week_of_year, day_of_week, is_weekend, price_rolling_avg, qty_rolling_avg, price_drop_pct, qty_spike_pct, promo_candidate

Calculation Rules — follow these exactly

- **Overall / aggregate Gross Profit %** = $\text{SUM}(\text{gross_profit}) / \text{SUM}(\text{sales})$ — always use this ratio-of-sums method for any period longer than one day (monthly, yearly, promo-period, overall).
- **Average Daily Gross Profit %** = $\text{MEAN}(\text{gross_profit_pct})$ — this is a different number from the above and answers a different question ("what does a typical day look like" vs. "what did the whole period actually earn"). Never conflate the two — state which one you're using when giving a GP% figure.
- **Weighted Average Unit Price** = $\text{SUM}(\text{sales}) / \text{SUM}(\text{quantity_sold})$. Simple mean of price_per_unit is a different, also-valid figure — label whichever you use.
- **Price Elasticity of Demand (PED)** = $\% \text{change in quantity} / \% \text{change in price}$, comparing a promotional window to its 30-day non-promotional baseline immediately before it.
- Gross Profit % and Gross Profit % per Unit are mathematically identical — don't present them as two separate insights.

Promotional Period Logic

- A day is promo_candidate = TRUE if it shows a price drop combined with a quantity spike relative to the trailing rolling average (already pre-computed in the data — don't try to re-derive this from raw price/quantity alone).
- When asked to analyze "a promotion" or "promotional periods," use promo_candidate to segment days, not manual price thresholds you invent.

- When comparing promo vs. baseline performance, use a 30-day non-promotional window immediately preceding the promo period unless the user specifies otherwise.

How to Answer

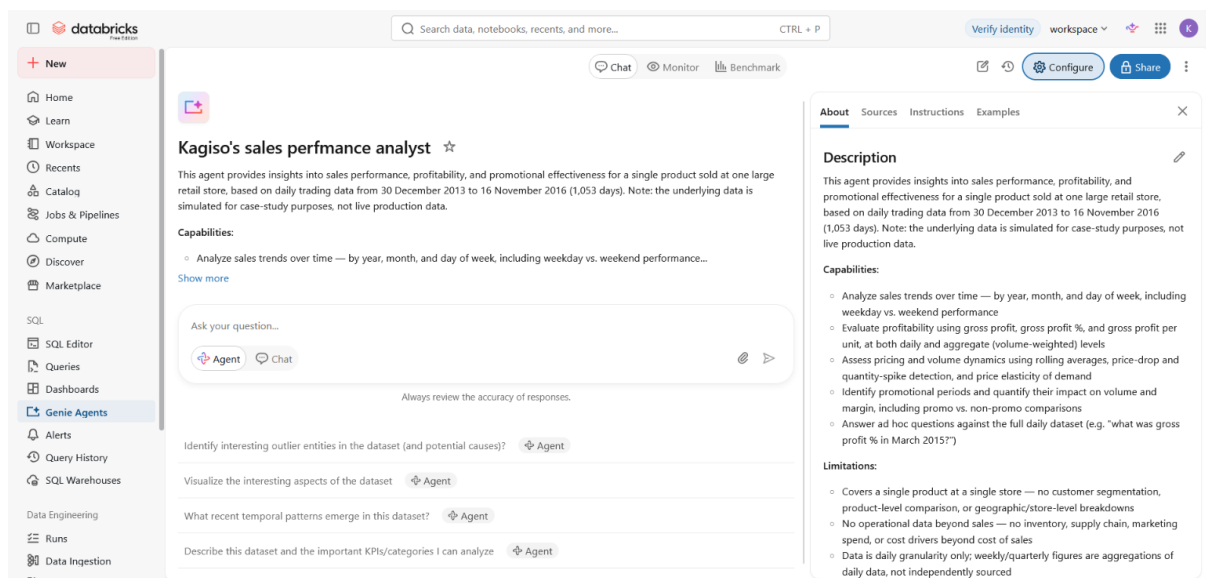
- Lead with the number, then the context — don't bury the answer in caveats.
- When a metric could be computed two valid ways (see Calculation Rules), state which one you used, briefly.
- Cite the actual date range or subset of days a figure is based on (e.g., "across all 1,053 days" or "for the 128 promotional days").
- If a question falls outside the data's scope (see Limitations below), say so plainly rather than guessing or extrapolating.
- Keep answers concise for simple factual questions; give a short structured breakdown for comparative or multi-part questions.
- Don't fabricate precision — if a figure is a rough estimate or visual read, say "approximately," not a false-precision decimal.

Guardrails

- Don't answer questions about other products, other stores, customer demographics, inventory, or marketing spend — this data doesn't contain that, and you should say so rather than speculating.
- Don't present the simulated data's findings as predictions or guarantees about real-world retail behavior — frame insights as "in this dataset" or "based on this case study."

- If asked to recommend a business action (e.g., "should we discount deeper?"), ground the recommendation explicitly in the data pattern that supports it (e.g., elasticity figures), and note it's a data-driven suggestion, not a guaranteed outcome.

Step 6 : Sales data agent was fully created and ready to be tested



Step 7 : Testing and validating the agent

i) Question :

What is the total sales, total quantity sold, overall gross profit %, and volume-weighted average unit price across the full dataset?

Answer :

Should return: R186.9m, 5,279,872 units, -3.81%, R35.40.

Result from Agent :

The screenshot shows the Databricks Genie Agents interface. The left sidebar contains navigation options: Home, Learn, Workspace, Recents, Catalog, Jobs & Pipelines, Compute, Discover, Marketplace, SQL, SQL Editor, Queries, Dashboards, Genie Agents (selected), Alerts, Query History, SQL Warehouses, Data Engineering, Runs, and Data Ingestion. The main area displays a query result for the question: "What is the total sales, total quantity sold, overall gross profit %, and volume-weighted average unit price across the full dataset?". The result is summarized in a table:

Metric	Value
Total Sales	R186,909,828.26
Total Quantity Sold	5,279,872 units
Overall Gross Profit %	-3.81% (a net loss)
Volume-Weighted Average Unit Price	R35.40

Below the table, a note states: "The negative gross profit percentage indicates that across the entire case study period, cost of sales exceeded gross profit, resulting in an overall loss on this product line. 1". The interface also includes a search bar, a chat input field, and a "Share" button.

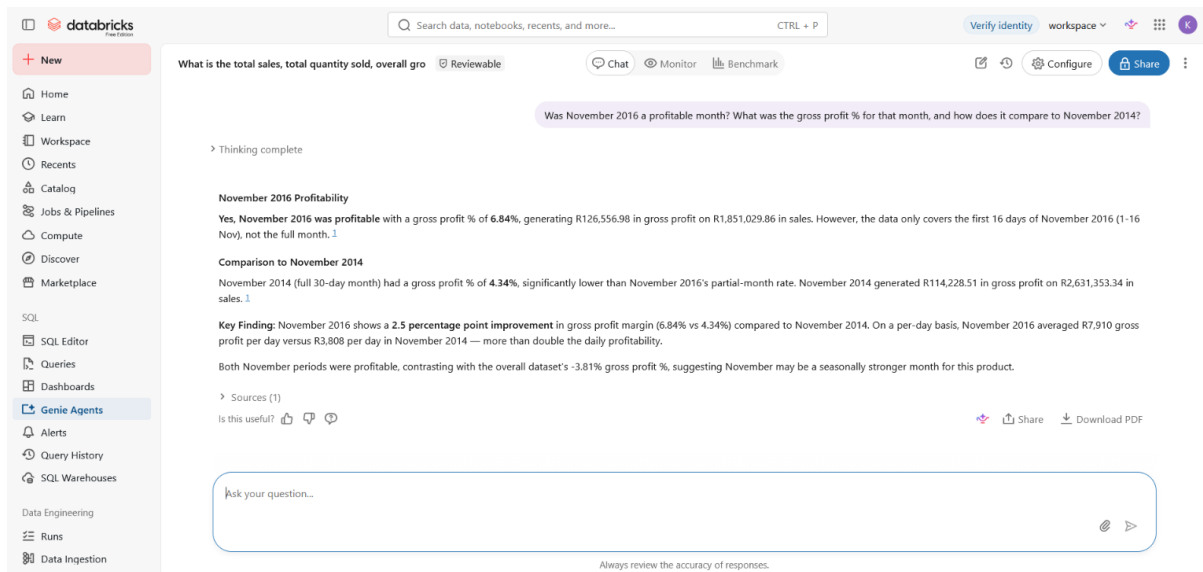
ii) Question :

Was November 2016 a profitable month? What was the gross profit % for that month, and how does it compare to November 2014?

Answer:

Should return: Nov 2016 $\approx$ +6.8% (profitable), Nov 2014 $\approx$ +4.3%

Result :



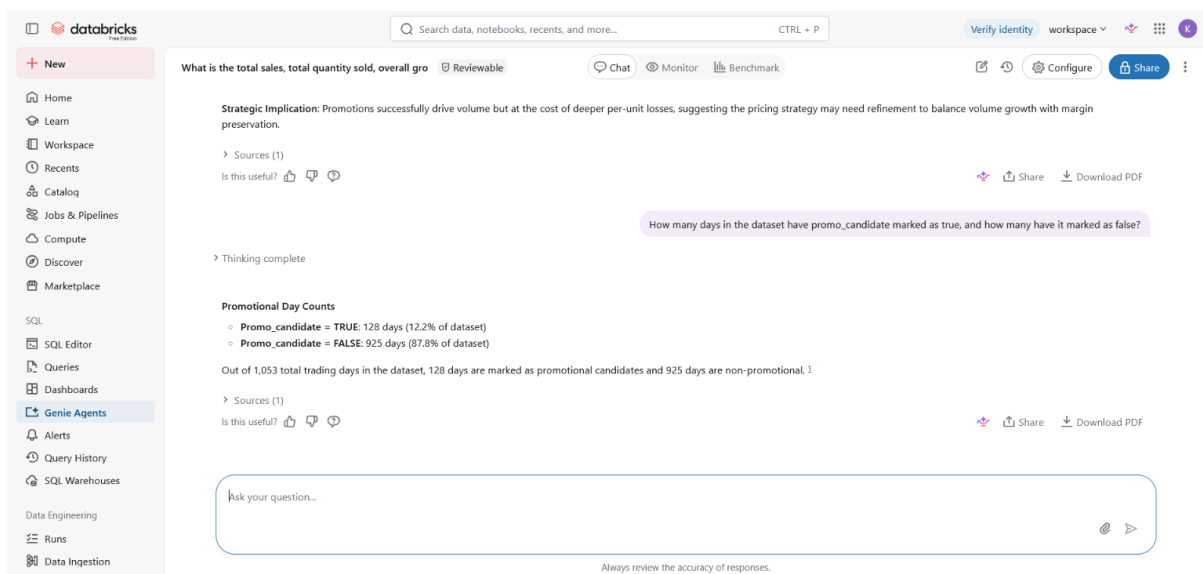
iii)Question:

How many days in the dataset have promo_candidate marked as true, and how many have it marked as false?

Answer:

Should return: 128 true, 925 false.

Result :



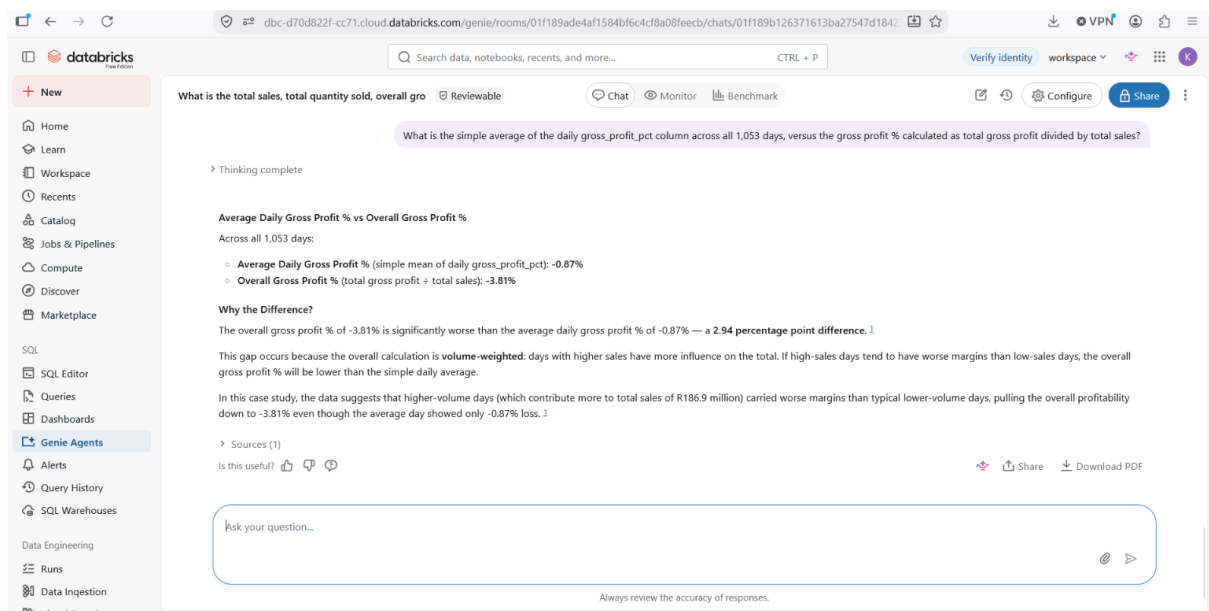
iv)Question:

What is the simple average of the daily gross_profit_pct column across all 1,053 days, versus the gross profit % calculated as total gross profit divided by total sales?

Answer:

Should return: $\approx -0.87\%$ (mean of daily column) vs $\approx -3.81\%$ (ratio of sums)

Result :



The screenshot shows the Databricks Genie interface. The question is: "What is the simple average of the daily gross_profit_pct column across all 1,053 days, versus the gross profit % calculated as total gross profit divided by total sales?". The response is as follows:

Thinking complete

Average Daily Gross Profit % vs Overall Gross Profit %

Across all 1,053 days:

- **Average Daily Gross Profit %** (simple mean of daily gross_profit_pct): -0.87%
- **Overall Gross Profit %** (total gross profit ÷ total sales): -3.81%

Why the Difference?

The overall gross profit % of -3.81% is significantly worse than the average daily gross profit % of -0.87% — a **2.94 percentage point difference**.¹

This gap occurs because the overall calculation is **volume-weighted**: days with higher sales have more influence on the total. If high-sales days tend to have worse margins than low-sales days, the overall gross profit % will be lower than the simple daily average.

In this case study, the data suggests that higher-volume days (which contribute more to total sales of R186.9 million) carried worse margins than typical lower-volume days, pulling the overall profitability down to -3.81% even though the average day showed only -0.87% loss.¹

Sources (1)

Is this useful?

Ask your question...

Always review the accuracy of responses.

The Agent answers most of the questions correctly , returning the answers I anticipated and also it matches with my findings in the presentation as well so the agent can be of great assistance when doing some analysis.

